

ERDENESANT, Mongolia

WMO: 442860

Lat: 47.3514N

Lon: 104.4917E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.0	(c) -27.9	(d) -33.9	(e) 0.2	(f) -29.4	(g) -32.0	(h) 0.2	(i) -27.3	(j) 10.3	(k) -17.1	(l) 9.7	(m) -16.0	(n) 1.8	(o) 180	(p) 0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 30.0	(d) 14.3	(e) 28.0	(f) 14.0	(g) 26.2	(h) 13.4	(i) 17.1	(j) 25.1	(k) 16.1	(l) 23.5	(m) 15.2	(n) 22.5	(o) 4.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 14.8	(b) 13.7	(c) 19.6	(d) 13.6	(e) 12.7	(f) 18.8	(g) 12.6	(h) 11.9	(i) 18.3	(j) 57.2	(k) 24.8	(l) 53.6	(m) 23.5	(n) 50.6	(o) 22.4	(p) 21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.1	(b) 9.7	(c) 8.5	(e) -32.6	(f) 32.8	(g) 2.4	(h) 2.1	(i) -34.3	(j) 34.3	(k) -35.7	(l) 35.6	(m) -37.0	(n) 36.8	(o) -38.7	(p) 38.4		
(a) 11.1	(b) 9.7	(c) 8.5	(e) -32.8	(f) 18.5	(g) 2.3	(h) 1.4	(i) -34.5	(j) 19.5	(k) -35.8	(l) 20.3	(m) -37.2	(n) 21.1	(o) -38.9	(p) 22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-18.1	-15.2	-4.9	4.0	10.3	16.2	18.9	16.7	10.9	2.4	-8.3	-15.5	(6)
(7)		DBStd	14.01	5.38	6.36	6.57	5.67	5.58	4.44	3.57	3.73	5.08	5.11	6.41	5.63	(7)
(8)		HDD10.0	3934	872	707	463	191	68	4	0	2	48	238	549	791	(8)
(9)		HDD18.3	6249	1131	940	722	430	255	92	37	76	226	493	799	1049	(9)
(10)		CDD10.0	839	0	0	0	11	76	189	274	209	76	3	0	0	(10)
(11)		CDD18.3	112	0	0	0	0	5	27	53	25	3	0	0	0	(11)
(12)		CDH23.3	1216	0	0	0	1	82	328	489	269	46	0	0	0	(12)
(13)		CDH26.7	331	0	0	0	0	16	93	148	67	7	0	0	0	(13)
(14)	Wind	WSAvg	3.6	3.4	3.4	3.8	4.4	4.7	3.8	3.1	3.0	3.3	3.5	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	231	2	1	4	6	18	41	66	58	20	7	4	4	(15)
(16)		PrecMax	357	6	5	12	16	46	88	114	98	58	20	17	15	(16)
(17)		PrecMin	113	0	0	0	1	3	11	19	14	3	2	0	1	(17)
(18)		PrecStd	72	2	2	3	4	12	20	31	31	15	4	4	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	3.7	13.8	21.8	29.2	32.8	33.3	30.9	27.6	18.7	10.2	0.0	(19)
(20)			MCWB	-5.8	-1.8	3.8	7.6	12.5	14.9	16.1	15.4	12.0	6.8	2.6	-3.7	(20)
(21)		2%	DB	-5.2	0.1	10.0	19.0	25.9	29.2	30.3	28.4	24.4	15.8	6.7	-3.3	(21)
(22)			MCWB	-8.0	-4.2	1.8	6.5	10.4	13.7	15.0	14.3	11.2	5.6	0.5	-6.1	(22)
(23)		5%	DB	-7.6	-2.7	7.2	16.4	23.4	26.9	28.0	26.4	22.1	13.3	3.7	-5.6	(23)
(24)			MCWB	-9.9	-6.0	0.5	5.4	9.6	12.7	14.7	13.8	10.5	4.5	-1.3	-7.8	(24)
(25)		10%	DB	-10.2	-5.4	4.7	13.9	20.6	24.9	26.0	24.2	19.7	10.8	0.8	-7.7	(25)
(26)			MCWB	-12.0	-7.8	-0.8	4.3	8.4	12.1	14.6	13.4	9.2	3.3	-3.2	-9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.6	-1.8	4.8	9.1	13.3	16.6	19.2	18.2	13.5	7.5	2.8	-3.6	(27)
(28)			MCDWB	-2.1	3.5	13.2	20.2	26.7	27.8	25.5	27.3	23.4	17.2	9.6	0.0	(28)
(29)		2%	WB	-8.0	-4.1	2.3	7.0	11.6	15.3	17.6	16.5	12.2	5.9	0.7	-6.0	(29)
(30)			MCDWB	-5.4	0.0	9.1	17.6	23.5	25.0	25.1	23.3	20.8	15.0	6.2	-3.3	(30)
(31)		5%	WB	-10.0	-5.9	0.7	5.8	10.1	14.4	16.7	15.4	11.2	4.7	-1.2	-7.8	(31)
(32)			MCDWB	-7.7	-2.8	6.9	15.5	21.5	23.3	24.3	21.9	19.9	12.6	3.2	-5.7	(32)
(33)		10%	WB	-12.0	-7.8	-0.8	4.6	8.8	13.5	15.8	14.6	10.2	3.4	-2.9	-9.5	(33)
(34)			MCDWB	-10.3	-5.3	4.3	13.4	20.1	21.6	23.0	21.1	18.5	10.4	0.6	-7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	10.8	11.3	11.8	12.5	11.5	10.3	10.8	11.9	11.5	9.7	8.5	(35)
(36)			MCDBR	11.8	13.4	13.3	15.4	15.4	14.5	13.7	14.3	14.6	14.7	12.3	10.2	(36)
(37)			MCWBR	9.7	10.3	7.9	7.6	7.0	5.9	5.1	5.6	6.2	8.0	7.9	8.3	(37)
(38)		5% WB	MCDBR	11.6	13.0	13.0	14.5	14.2	12.1	11.1	11.1	13.3	14.0	11.6	10.1	(38)
(39)			MCWBR	9.6	10.2	7.9	7.5	7.1	5.4	5.0	5.3	6.1	7.8	7.7	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.210	0.229	0.264	0.298	0.319	0.331	0.321	0.304	0.269	0.248	0.226	0.212	(40)	
(41)		taud	2.424	2.412	2.390	2.360	2.336	2.387	2.460	2.486	2.533	2.529	2.471	2.417	(41)	
(42)		Ebn at Noon	924	968	970	958	944	931	935	941	955	932	893	877	(42)	
(43)		Edh at Noon	63	81	98	113	121	116	106	99	85	72	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.91	3.05	4.53	5.77	6.74	6.76	6.42	5.64	4.78	3.28	2.03	1.53	(44)	
(45)		RadStd	0.15	0.20	0.23	0.23	0.38	0.40	0.32	0.34	0.20	0.17	0.16	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (5 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) -2.80	(g) N/A	(h) -0.66	(i) -0.88	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page